

Content Refresh Opportunity Scoring: A Machine Learning Approach for Prioritizing Content Updates

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Abstract

Content refresh decisions are important for maintaining search visibility and user engagement across large content portfolios. This project investigates whether machine learning can support the prioritization of content updates using search and engagement signals from the FlyRank ML Internship dataset. A refresh opportunity scoring framework was developed using a rule-based baseline and a Random Forest Regressor model. The model was trained on content freshness, visibility, engagement, and search-performance features and evaluated using an 80/20 train-test split. Results indicate that content visibility, update recency, search position, and content age are among the strongest indicators of refresh opportunity, providing a practical decision-support system for editorial teams.

1 Introduction

Managing large collections of content presents a significant challenge for organizations that depend on organic search traffic. As content ages, information may become outdated, rankings may decline, and engagement may decrease. Identifying which pages require attention is often a manual and time-consuming process.

This project addresses the problem of prioritizing content updates through a data-driven refresh opportunity scoring system. The goal is not to predict future search performance directly but to provide an evidence-based ranking of pages that may benefit from editorial review.

The work supports editorial decision-making by helping teams allocate resources more efficiently and identify high-priority content refresh opportunities.

2 Problem Statement

The objective of this project is to develop a machine learning-based system that ranks content pages according to their refresh opportunity.

Unit of Analysis: One content page represented by a single row in the dataset.

Output: A Refresh Opportunity Score used to prioritize editorial review.

Decision Supported: Which pages should be reviewed, refreshed, monitored, or maintained.

Business Value: Efficient allocation of editorial resources and systematic identification of high-priority content updates.

3 Dataset Description

This study uses the FlyRank ML Internship dataset.

3.1 Dataset Summary

- Total Pages: 30,000
- Total Features: 44
- Content-Level Search and Engagement Data
- Anonymized Client and Content Identifiers

3.2 Included Features

The analysis uses search, content, and engagement-related variables including:

- Search Volume
- Competition
- CPC
- Word Count
- Character Count
- Impressions
- Clicks
- Sessions
- Content Age
- Days Since Last Update
- CTR
- Average Position
- Engagement Rate
- Scroll Rate

3.3 Excluded Features

The following variables were excluded from modeling:

- `content_id`
- `client_id`
- `trend_direction`
- `trend_pct`

These fields were excluded to avoid information leakage and prevent the model from learning directly from trend-related outcomes.

4 Methodology

4.1 Assumptions

This project assumes that content freshness, search visibility, and engagement metrics provide useful signals for identifying content refresh opportunities. Pages that are older, less visible, or experiencing declining engagement may benefit from editorial review.

4.2 Target Definition

The target variable is a Refresh Opportunity Score.

Rather than predicting a binary outcome, the system ranks pages according to their estimated need for content updates.

4.3 Baseline Scoring System

A transparent rule-based baseline was created using three signals:

- Content Staleness (40%)
- Search Demand (30%)
- Declining Trend Signal (30%)

The baseline score was calculated as:

$$Score = 0.40(Staleness) + 0.30(SearchDemand) + 0.30(TrendSignal)$$

This baseline serves as the reference point for model evaluation.

4.4 Machine Learning Model

A Random Forest Regressor was selected because of its ability to model nonlinear relationships and provide feature importance estimates.

Model Configuration:

- Algorithm: Random Forest Regressor
- Number of Trees: 100
- Random State: 42
- Train/Test Split: 80/20

5 Results

5.1 Model Performance

The model was evaluated on the test set using three standard regression metrics.

Metric	Value
MAE	11.49
RMSE	13.35
R ²	0.298

Table 1: Random Forest Performance Metrics

The model captures approximately 29.8% of the variance in the refresh opportunity score. While performance is moderate, the results indicate that meaningful patterns exist within the available features.

5.2 Feature Importance Analysis

Feature importance analysis revealed that the model relied most heavily on content visibility and freshness-related signals.

Top Influential Features:

1. impressions_90d
2. days_since_last_update
3. avg_position
4. content_age_days
5. char_count
6. word_count
7. sessions_90d
8. scroll_rate

9. ctr
10. clicks_90d

These findings align with editorial expectations and support the validity of the scoring framework.

6 Discussion

The model identified content visibility, update recency, search position, and content age as the most important signals for refresh prioritization.

Pages that have not been updated recently and show signs of weaker search performance tend to receive higher refresh scores. This supports the practical intuition that aging content requires periodic review to remain competitive.

The results also demonstrate that machine learning can assist editorial teams by synthesizing multiple performance indicators into a single prioritization framework.

7 Limitations

Several limitations should be considered.

First, the target variable is a proxy score rather than a directly observed measure of refresh success. The model therefore learns relationships associated with the scoring framework rather than confirmed outcomes.

Second, trend-related variables were intentionally excluded to avoid leakage. While this improves methodological validity, it also reduces predictive performance.

Third, the analysis is observational and does not establish causal relationships. A high refresh score does not guarantee that updating a page will improve rankings, traffic, or engagement.

The findings should therefore be interpreted as directional and decision-support evidence rather than definitive predictions.

8 Ranked Recommendations

Based on the scoring framework and model outputs, the following editorial actions are recommended.

8.1 Priority 1: Immediate Refresh Review

Pages with the highest refresh opportunity scores should be reviewed immediately. These pages typically combine content staleness, declining performance, and strong search demand.

8.2 Priority 2: Monitor and Improve

Pages with moderate refresh scores should be monitored and periodically updated as needed.

8.3 Priority 3: Maintain Current Performance

Pages with low refresh scores should remain under observation but generally require limited intervention.

9 Reproducibility

Repository:

<https://github.com/ihaseeb0081/flyrank-ml-internship-haseeb>

Notebook:

<https://github.com/ihaseeb0081/flyrank-ml-internship-haseeb/blob/main/work/notebooks/capstone.ipynb>

Model Configuration:

- Random State: 42
- Train/Test Split: 80/20
- Random Forest Trees: 100

The project can be reproduced by cloning the repository, loading the FlyRank dataset, executing the notebook, and running all cells sequentially.

10 Future Work

Future improvements may include:

- Time-aware validation strategies
- Advanced ranking models
- Learning-to-rank approaches
- Additional content quality features
- Editorial feedback loops

11 Acknowledgments

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Built using the FlyRank ML Internship Dataset.

FlyRank Website:

<https://flyrank.ai>

12 References

1. FlyRank ML Internship Dataset
2. Scikit-Learn Documentation
3. Random Forest Regression Literature
4. Search Intelligence and Content Optimization Research